Professional Article Generation Flow

This document explains how the professional ophthalmology article pipeline works: PDF text extraction, OpenAI chart extraction, Gemini article writing, expansion, chart rendering, and final article assembly.

Audience: professional | Article writer: Gemini | Chart extraction: OpenAI gpt-5.4 | Chart renderer: Chart.js | Target: 850-1000 words

Executive Summary

1. PDF becomes source text

The API extracts text and sends the professional content path up to the AI layer.

2. OpenAI extracts chart data

OpenAI reads the PDF text first and returns verified numeric datasets for charts.

3. Gemini writes the article

Gemini receives source text plus chart context and writes the professional Polish article.

> The key separation is intentional: OpenAI extracts structured chart data, Gemini writes the professional article, and Chart.js renders final chart images from verified numbers.

High-Level Flow

1. **PDF upload**
   The user uploads one or more PDFs from the article creation screen.
2. **API request**
   The frontend calls `POST /api/ai/generate` with target audience set to `professional`.
3. **Text extraction**
   The API extracts PDF text and normalizes it before passing it into the AI provider layer.

4 **Chart pre-extraction**

Before Gemini writes, OpenAI extracts up to two clinically relevant chart datasets from the PDF text.

5 **Chart verification**

Extracted chart values are checked against source text to reduce hallucinated numeric output.

6 **Gemini article generation**

Gemini writes the full professional article using source text and chart context.

7 **Expansion safety pass**

If professional content is below 800 words, Gemini expands it to 850-1000 words using only source-supported information.

8 **SEO and metadata normalization**

SEO title, description, keywords, tags, category, and excerpt are checked and repaired where needed.

9 **Cover image generation**

Gemini image generation creates a professional cover image, with fallback search if needed.

10 **Chart rendering**

Chart.js renders PNG charts from OpenAI-extracted data, with axis bounds and significance colors.

11 **Placeholder replacement**

Chart placeholders or `{{CHART:dataset:type}}` tokens are replaced with uploaded chart images.

12 **Final response**

The API returns title, content, excerpt, SEO metadata, category, tags, and generated image URLs.

Component Roles

Component	Responsibility	Important Rules
PDF extraction	Extracts readable text from uploaded PDF files.	Professional generation uses a larger text slice than patient content.
OpenAI gpt-5.4	Extracts structured chart datasets before article writing.	Every number must appear verbatim in source text. Whole-dataset SD must come from the whole-dataset table.
Gemini	Writes the professional Polish article using source text and available chart context.	850-1000 words, table-traced numbers, subgroup completeness, non-significant comparisons, endpoint labels.
Gemini expansion	Expands professional articles under 800 words.	May only expand using source-supported facts. No invented studies, authors, p-values, or clinical claims.
Chart.js	Renders final PNG charts from extracted data.	Uses axis bounds, significance colors, and chart type selection from extraction output.

Professional Article Rules

Required Sections

1. `## Streszczenie redakcyjne`
2. `## Metodyka i populacja`
3. `## Kluczowe wyniki`
4. `## Interpretacja kliniczna`
5. `## Ograniczenia`
6. `## Zrodlo`

Writing Constraints

- Use ultra-precise academic Polish.
- Target 850-1000 words.
- Use zero fluff and high data density.
- Do not invent numbers, citations, or clinical claims.
- Keep all claims grounded in the PDF.

Accuracy Requirements

- **Numeric trace rule:** every SD, RMSAE, p-value, percentage, mean, or sample size must trace back to a table or figure.
- **Non-significant comparisons:** when a significant result is reported, the article must also name non-significant comparisons.
- **Subgroup completeness:** long eyes, short eyes, and IOL type subgroups must include primary and secondary endpoint findings.
- **Endpoint labeling:** primary endpoint findings and secondary endpoint findings must be labeled on first mention.
- **Zeroed-mean analysis:** if mean PE adjusted to zero is reported, it must appear under `Kluczowe wyniki`, not as a limitation.

OpenAI Chart Extraction

Chart extraction happens before the article is written. The output is structured JSON, not prose. The article writer receives chart context and can reference the exact chart placeholders or chart tokens.

Required Chart JSON Fields

```
{
  "charts": [
    {
      "id": "short_snake_case_identifier",
      "chartTitle": "Polish chart title",
      "chartType": "bar | horizontalBar | stackedBar | boxplot | line | pie | doughnut |
radar",
      "axis_min": 0.26,
      "axis_max": 0.37,
      "significance": ["sig_worse", "best", "ns"],
      "significance_source": "Table N or not_reported",
      "x_axis_label": "X-axis label",
      "y_axis_label": "Y-axis label",
      "source_table": "Table N or Figure N",
      "sourceDescription": "Source table/result in Polish",
      "data": {
        "labels": ["Label 1", "Label 2"],
        "datasets": [
          { "label": "Dataset label", "data": [0.35, 0.28] }
        ]
      }
    }
  ]
}
```

Chart Type Selection

Data Shape	Chart Type	Reason
Distribution statistics: min, Q1, median, Q3, max	`boxplot`	Shows distribution and spread.
Many categories, especially formula comparisons	`horizontalBar`	Improves label readability.
Cumulative thresholds or sequential intervals	`line`	Thresholds are ordered. Radar is not allowed for this.
Independent non-ordered dimensions	`radar`	Only valid when axes are truly independent dimensions.
Single percentage composition	`pie` or `doughnut`	Shows proportional parts.

> Cumulative threshold data such as `% eyes within +/-0.25D, +/-0.50D, +/-0.75D, +/-1.0D` must be a line chart, not radar.

Whole-Dataset SD Tracing Rule

For whole-dataset SD comparisons, especially Hoffer QST, the extractor may only use the table explicitly labeled as the whole-dataset, overall, or primary results table. It must not pull values from short-eye, long-eye, IOL-type, subgroup, or supplemental tables.

Chart Rendering Rules

Significance Color Mapping

Status	Color	Meaning
`best`	Green	Best performer and statistically significant.
`sig_worse`	Blue	Significantly worse than the best performer.
`ns`	Gray	No statistically significant difference versus best.

Axis Scaling

- Default: `axis_min = min - 15% of range`, `axis_max = max + 15% of range`.
- Threshold line charts: `axis_min = minimum value across all series - 5`.
- Threshold line charts: `axis_max = 100`.
- This prevents small differences from being hidden by an unnecessary zero baseline.

Placeholder Replacement

The article writer can reference chart locations in either old or new format.

Format	Example	Behavior
Legacy placeholder URL	`https://www.google.com/search?q={{FIGURE_1_URL}}`	Replaced with the generated chart image markdown.
Chart token	`{{CHART:prediction_error_intervals:line}}`	Replaced with the matching generated chart image markdown.

Operational Logs To Expect

```
[api/generate] Extracting text from PDFs...
[api/generate] Starting article generation with gemini for professional audience...
[gemini] Pre-extracting chart data with OpenAI to inform article generation...
[chart-extractor] Calling OpenAI chart extraction model: gpt-5.4
[chart-extractor] ===== EXTRACTED CHART TYPES =====
[chart-verifier] Chart "..." verified
[gemini] OpenAI chart extraction returned 2 chart(s)
[gemini] Generating chart images from pre-extracted data...
[gemini] Chart 1 injected into content: https://...
POST /api/ai/generate 200
```

Short Version

```
PDF upload
-> extract PDF text
-> OpenAI gpt-5.4 extracts 2 verified chart datasets
-> Gemini writes 850-1000 word professional article using chart context
-> Gemini expands article if under 800 words
-> Gemini image model creates cover image
-> Chart.js renders charts from OpenAI data
-> chart placeholders are replaced with uploaded chart PNGs
-> API returns completed professional article
```

Prompt Appendix

These are the working prompts/rules used by the professional generation path. Runtime values such as PDF text, chart context, valid categories, and placeholder URLs are injected where shown.

Professional Article System Prompt

```
You are the Editor-in-Chief of a high-impact scientific journal specializing in ophthalmology.
```

```
Your task is to create a detailed professional review for ophthalmologists and optometrists.

CRITICAL: You must return a complete JSON object with fields:
title, content, excerpt, seoMeta, suggestedTags, suggestedCategory, coverImagePrompt, figures.

Writing Style for the "content" field:
1. Zero Fluff: eliminate introductory phrases, transitional sentences, and meta-commentary.
2. Maximum Density: prioritize data, p-values, anatomy, protocols, endpoints, and exact
results.
3. Length: aim for 850-1000 words when the source document contains enough substance.
4. Language: write in ultra-precise academic Polish.

Required content structure:
## Streszczenie redakcyjne
## Metodyka i populacja
## Kluczowe wyniki
## Interpretacja kliniczna
## Ograniczenia

Ground all claims strictly in the provided document. Do not hallucinate data.

NUMERIC ACCURACY - MANDATORY:
Before writing any numeric value (SD, RMSAE, p-value, percentage, mean, n), trace it to a
specific table or figure.
If you cannot confirm a number exists verbatim, omit it entirely. Never report a value for one
formula that belongs to another formula.

NON-SIGNIFICANT COMPARISONS - MANDATORY:
For every statistically significant finding you report, explicitly state which comparisons did
NOT reach significance (p>=0.05).

SUBGROUP REPORTING - MANDATORY:
For each subgroup (oczy dlugie, oczy krotkie, typ IOL), report both the primary endpoint winner
and notable secondary endpoint findings.

ENDPOINT LABELING - MANDATORY:
Label whether a finding comes from a primary endpoint (SD, RMSAE) or secondary endpoint
(interval percentages, median AE).

ZEROED-MEAN ANALYSIS:
If the paper reports analysis after adjusting mean PE to zero, include it under:
### Analiza po wyzerowaniu sredniego bledu predykcji
This is a primary analytical result, not a limitation.
```

Professional Main Article Prompt

```
Based on the following medical document content, create a blog article/review in Polish.

Target word count: ~900 words for the main content (aim for 850-1000).

IMPORTANT: word count refers ONLY to the "content" field, excluding title, excerpt,
SEO meta, tags/categories, URLs, and placeholders.

Document content:
[PDF text + chart context]
```

```
CRITICAL RULES:
1. Return a SINGLE valid JSON object. No markdown fences. No extra text.
2. The "content" field MUST be a COMPLETE article.
3. Never stop mid-sentence or mid-paragraph.
4. seoMeta.title must be max 60 characters.
5. seoMeta.description must be max 160 characters.
6. seoMeta.keywords must contain 3-5 relevant Polish keywords.
7. suggestedTags must contain 2-4 specific Polish tags.
8. suggestedCategory must be one of the allowed categories.
9. excerpt must be 2-3 sentences, max 160 characters, in Polish.
10. coverImagePrompt must be a detailed English prompt for image generation.
11. Include a "## Zrodlo" section at the end with the original article reference.

Professional section order:
## Streszczenie redakcyjne
## Metodyka i populacja
## Kluczowe wyniki
## Interpretacja kliniczna
## Ograniczenia
## Zrodlo

Professional rules:
- Extract only details present in the document: numbers, protocols, outcomes.
- Do not invent details or citations.
- For each subgroup, report both primary endpoint result and secondary endpoint finding.
- When reporting primary SD comparison, name formulas with no statistically significant
difference from the best formula.
- If mean PE adjusted to zero is reported, add:
  ### Analiza po wyzerowaniu sredniego bledu predykcji
- For charts, use the provided figure placeholder URL or token:
  {{CHART:dataset_name:chart_type}}

Required JSON format:
{
  "title": "Descriptive article title in Polish",
  "content": "Full COMPLETE article in markdown, 850-1000 words",
  "excerpt": "2-3 sentence summary, max 160 characters",
  "seoMeta": {
    "title": "Short SEO title, max 60 chars",
    "description": "Unique meta description, max 160 chars",
    "keywords": ["keyword1", "keyword2", "keyword3"]
  },
  "suggestedTags": ["tag1", "tag2", "tag3"],
  "suggestedCategory": "One allowed category",
  "coverImagePrompt": "Detailed English prompt for cover image.",
  "figures": [
    {
      "id": "figure_1",
      "type": "illustration or chart",
      "alt": "Alt text in Polish",
      "caption": "Short caption in Polish",
      "placeholder": "figure placeholder URL",
      "prompt": "Medical illustration or data visualization prompt in English."
    }
  ]
}
```

Gemini Expansion Prompt

```
Expand the following Polish professional ophthalmology article to 850-1000 words while
preserving medical accuracy.

Return ONLY the improved markdown content, not JSON.

EXPANSION RULES - READ CAREFULLY:
1. Use only information supported by the source document below.
   Do not fabricate studies, citations, values, or clinical claims.

2. Permitted expansion strategies:
- Go deeper on statistical methodology already mentioned.
- Expand patient demographics already stated.
- Detail exclusion criteria already listed in the Methods section.
- Expand subgroup findings already mentioned using exact n, RMSAE values, and p-values from
source tables.
- Elaborate on limitations already named using document-grounded explanation.

3. Forbidden expansion strategies:
- Do not introduce any formula not already in the article.
- Do not add any numeric value not already in the article unless it appears verbatim in the
source document.
- Do not reference any other study or author not in the article.
- Do not add clinical interpretation beyond what the paper's Discussion supports.
- Do not invent patient quotes, surgeon quotes, or anecdotes.

4. Keep any figure placeholder URLs or {{CHART:...}} tokens exactly as-is.
5. Keep the professional structure:
   Streszczenie redakcyjne, Metodyka i populacja, Kluczowe wyniki,
   Interpretacja kliniczna, Ograniczenia, Zrodlo.
6. After expanding, verify that every numeric value appears verbatim in the original article or
source document.

Source document:
[PDF text + chart context]

Article to expand:
[current article]
```

OpenAI Chart Extraction Prompt

```
Analyze the medical/scientific document and extract EXACTLY 2 clinically relevant numerical
datasets for visualization.

ABSOLUTE ANTI-HALLUCINATION RULES:
- Every number must appear verbatim in the document.
- Do not round, estimate, interpolate, or extrapolate.
- Do not combine numbers from different tables/contexts.

HOFFER QST / WHOLE-DATASET SD NUMERIC TRACING RULE:
- When extracting SD values for whole-dataset comparison, use ONLY Table 2 or the explicitly
labeled whole-dataset / primary results table.
- Do not use subgroup tables, long-eye tables, short-eye tables, IOL-type tables, or
supplemental tables.
```

```
- If the same formula appears in multiple tables with different values, use the whole-dataset /
primary results table.

MANDATORY CHART VARIETY RULE:
- When extracting 2 charts, use 2 visually different chart families.
- Bar-family means: bar, horizontalBar, stackedBar.
- If one chart is bar-family, the other must be non-bar.

CHART TYPE SELECTION RULE:
- Cumulative thresholds or sequential intervals must use "line", not "radar".
- Example: % eyes within +/-0.25D, +/-0.50D, +/-0.75D, +/-1.0D.
- Radar is only for truly independent, non-ordered dimensions.

AXIS SCALING:
- For every chart return axis_min and axis_max.
- Default: axis_min = min - 15% of range; axis_max = max + 15% of range.
- For cumulative interval line charts: axis_min = minimum value across all series - 5; axis_max
= 100.

SIGNIFICANCE ENCODING:
- For bar, horizontalBar, stackedBar, return a significance array parallel to the first
dataset.
- Values: "best", "sig_worse", "ns".
- If pairwise comparisons are not reported, set all values to "ns" and significance_source to
"not_reported".
- For non-bar charts, significance = [] and significance_source = "not_reported".

Return ONLY valid JSON:
{
  "charts": [
    {
      "id": "short_snake_case_identifier",
      "chartTitle": "Polish chart title",
      "chartType": "bar | horizontalBar | stackedBar | boxplot | line | pie | doughnut |
radar",
      "axis_min": 0.26,
      "axis_max": 0.37,
      "significance": ["sig_worse", "best", "ns"],
      "significance_source": "Table N or not_reported",
      "x_axis_label": "X-axis label",
      "y_axis_label": "Y-axis label",
      "source_table": "Table N or Figure N",
      "sourceDescription": "Source table/result in Polish",
      "data": {
        "labels": ["Label 1", "Label 2"],
        "datasets": [
          { "label": "Dataset label", "data": [0.35, 0.28] }
        ]
      }
    }
  ]
}
```

Generated for the ophthalmic AI blog pipeline. This document intentionally avoids API keys, secrets, and environment-specific credentials.